SUPPORTING ADULT GUIDANCE

Where does the line cross the y-axis?

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner isolates and interprets the **y-intercept c** — the value of y when $x = 0$ — while the gradient is held fixed. The emphasis is on **controlled variation**: changing one thing (c) and seeing exactly what it does. They also learn to read c from any representation and meet **parallel lines** (same m, different c).

Driving Question: *What does c control when nothing else has changed?*

The mathematics, in plain terms

In $\mathbf{y = mx + c}$, the number **c** is the **y-intercept**: the value of y when $x = 0$. On a graph it is the height where the line crosses the **y-axis**, the point **(0, c)**. In an equation it is the number added on its own. In a table it is the y value in the $x = 0$ column. Because c only sets the height where the line crosses the y-axis, changing it **slides the whole line straight up or down** without tilting it — the steepness (the gradient m) is untouched. That is why two lines with the **same gradient but different intercepts** are **parallel**: same slope, different intercept, so they never meet. It is worth being precise: a line is not a journey with a beginning — it extends for ever both ways — so c is not the line's 'start' but simply where it meets the y-axis. In a real-life context where x counts from 0, that value is the **initial or fixed amount** — the standing charge before any usage, the deposit already saved, the call-out fee before any work. (Note that $y = mx + c$ describes only non-vertical lines; a vertical line $x = k$ has no y-intercept and cannot be written this way.) This lesson deliberately holds the gradient fixed so the effect of c can be seen cleanly on its own; the gradient gets the same treatment in the next lesson.

Change one thing at a time

The heart of this lesson is controlled variation: lock the gradient, move only c, and see precisely what c does. That discipline — changing one variable while holding the other fixed — is a powerful scientific habit, and it is what lets a learner attribute the sliding motion entirely to c. Protect the observation that the **steepness never changes**; it is the whole point, and it leads directly to why same-gradient lines are parallel. Keep the plain phrase 'the value when x = 0' alongside the formal term 'y-intercept', and avoid calling it the line's 'starting point' — a line has none. Once c is secure across a graph, an equation and a table, the learner can find the intercept of any line in any form — the groundwork for writing equations two lessons from now.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	Intercept Isolator: gradient locked, slide c and watch the line move up and down through (0, c).	~15 min
Movement	Slide to a Target: drag c until the line passes through a marked point on the y-axis.	~8 min
Reflection	Find the Intercept: read c from a graph, an equation and a table.	~10 min
Creativity	Parallel Family: build three lines with the same gradient and different starts.	~5 min
Rest	A pause after holding one thing fixed while changing another.	~4 min
Nutrition	Contextual (Mode A): standing charges, deposits, call-out fees — the fixed amount before anything happens.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking changing c also changes the steepness

Repair: “Slide c up and down — does the line tilt, or just move? What stays exactly the same?”

Reading the x-intercept instead of the y-intercept

Repair: “c is where the line meets the y-axis — the vertical axis, where $x = 0$. Which crossing is that?”

Missing a negative intercept

Repair: “If the line crosses below the origin, is c positive or negative? Read the y value there.”

Not seeing c in an equation with a subtraction

Repair: “In $y = 2x – 3$, the ‘– 3’ means $c = –3$. What is c in $y = 4x – 1$?”

Thinking parallel lines must be identical

Repair: “Same gradient makes them parallel; different c makes them separate. Can two different lines still be parallel?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading coordinates, and substituting $x = 0$ into a rule to find the y-intercept. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the single-variable focus — and the reading of c — with the learner:

- “Where does the line cross the y-axis? That height is c.”
- “Slide c. Did the steepness change, or only the position?”
- “What is c from this equation? And from this table (y when x = 0)?”
- “These two lines have the same gradient — what does that make them?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Isn't c the start of the line?”

Not quite — and this is worth getting right. A line has no start: it carries on for ever in both directions. c is simply the y-intercept, the value of y when x = 0, where the line crosses the y-axis. In a real situation where x counts from 0 (visits, weeks, minutes), that value happens to be the initial amount — which is why 'starting value' feels natural — but it is a property of the context, not of the line itself.

“Why hold the gradient fixed?”

So the learner can see the effect of c on its own. If both numbers change at once, it is hard to tell which is doing what; locking the gradient makes c's role unmistakable.

“Is the y-intercept always positive?”

No — c can be negative (the line crosses below the origin) or zero (through the origin). The tools let the learner slide c through all of these.

“How does this help with writing equations later?”

Being able to read c from any representation is half of writing y = mx + c. The next lesson isolates the gradient m in the same way; together they let a learner write the equation of any line.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.